

Second Terminal Examination - 2079

Class : XII

Subject: Chemistry (3021)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Circle the best alternatives.

[1 × 11 = 11]

- The average minimum energy that must be associated with the reactant molecules in order to be capable of forming product is,
a. Activation energy b. Collision energy
c. Threshold energy d. Transition energy
- How much water should be evaporated from 400 ml of $\frac{N}{10}$ HCl to make it exactly 2N?
a. 360 ml b. 370 ml c. 380 ml d. 390 ml
- Which of the following does not act as Bronsted – Lowry acid?
a. CH_3COO^- b. HCO_3^- c. HSO_3^- d. NH_4^+
- Copper sulphate is not stored in aluminium bottle because
a. Cu get oxidized b. Cu^{++} get reduced
c. Al^{+++} get reduced d. CuSO_4 get decomposed
- During the extraction of iron from hematite, the limestone acts as;
a. Reducing agent b. Gangue
c. Flux d. Stag

- An organic compound 'X' undergoes reduction with LiAlH_4 to yield 'Y'. When vapour of 'Y' are passed over heated copper at 300°C , 'X' is formed. The compound 'Y' is
 - $\text{CH}_3 - \text{CHO}$
 - $\text{CH}_3 - \text{CH}_2 - \text{OH}$
 - $\text{CH}_3 - \text{CO} - \text{CH}_3$
 - $\text{CH}_3 - \text{O} - \text{CH}_3$
- The order of reactivity of alkyl halides towards elimination reaction is
 - $3^\circ > 2^\circ > 1^\circ$
 - $2^\circ > 1^\circ > 3^\circ$
 - $3^\circ > 1^\circ > 2^\circ$
 - $1^\circ > 2^\circ > 3^\circ$
- Chlorobenzene on nitration produces
 - o-nitrobenzene
 - p-nitrobenzene
 - o & p - chloronitrobenzene
 - m-nitrobenzene
- When a 2° alkyl halide reacts with an alkoxide, the product is
 - hydrocarbon
 - ether
 - unsaturated hydrocarbon
 - alcohol
- Nitroethane is prepared by reaction of
 - Sodium nitrite and bromoethane
 - Potassium nitrite and iodoethane
 - Silver nitrite and iodomethane
 - Fuming nitric acid and ethane
- Phenol reacts with PCl_5 to give mainly
 - p - chlorophenol
 - Chlorobenzene
 - o- and p-chlorophenols
 - Triphenylphosphate

Group - B

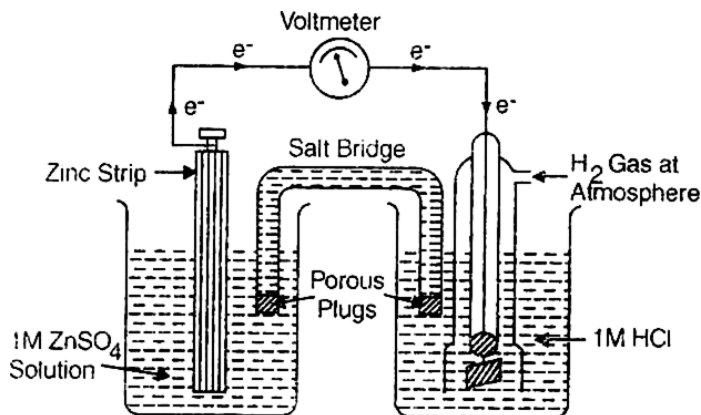
Give short answer to the following questions.

[8 × 5 = 40]

1. Concentration refers to amount of solute present in given volume of solution.
 - a. Define molarity and normality. [2]
 - b. Which one has higher concentration? 80 g/L NaOH solution and 3M NaOH solution [1]
 - c. X gm of Na_2CO_3 reacts completely with 20ml of 0.1M HCl. Calculate the value of x. [2]
2. G.N Lewis, in 1923, put forward an entirely new concept of acid and base from Arrhenius theory and Bronsted theory. This concept is known as Lewis concept of acid and base.
 - a. Write limitations of Bronsted-Lowry concept of acid and bases. [1]
 - b. Define Lewis acids and bases with examples. [2]
 - c. Define and illustrate acid- base reaction according to Lewis theory. [2]
3. Enzymes are high molecular weight protein molecules produced in the living cells of plants and animals, which possess the incredible capacity to catalyze i.e increases the rate of various complex biochemical reactions occurring in the body of the living system.
 - a. How an enzyme (Catalyst) does increase the rate of reaction? [1]
 - b. Write any two characteristics of enzyme catalysis. [1]
 - c. Give example of homogeneous and heterogeneous catalysis. [1]
 - d. Draw energy profile diagram for a catalyzed and uncatalyzed reaction. [2]

OR

The figure given below shows a redox couple containing Zn^{++}/Zn and standard hydrogen electrode (SHE). SHE is used to determine reduction potential of Zn^{++}/Zn electrode.



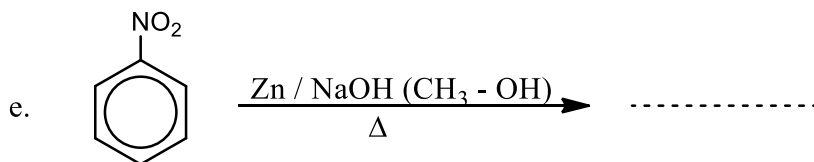
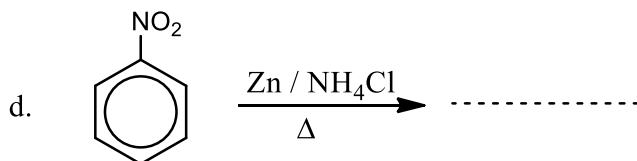
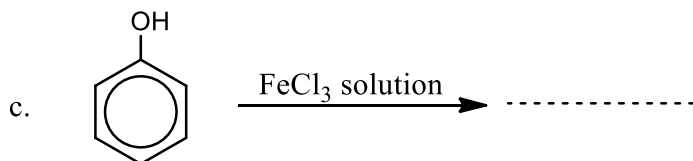
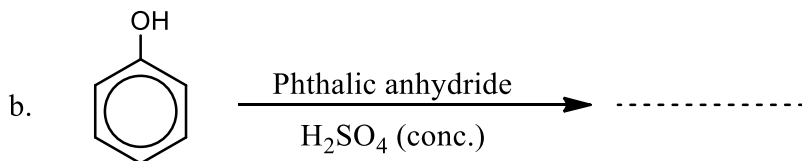
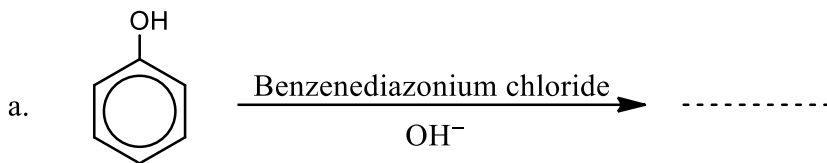
- a. What is SHE? Why is SHE called primary reference electrode? [2]
 - b. Define single electrode potential. Mention factors affecting the value of reduction potential. [2]
 - c. Write the applications of salt bridge. [1]
4. a. The crystal field theory (CFT) was developed for crystalline solids by the physicist Hans Bethe in 1929.
- i. Explain Crystal field theory for splitting of d-orbitals in octahedral complexes. [2]
 - ii. Write the possible oxidation states for Cr and Mn. [1]
 - iii. Compounds of Sc^{+2} are coloured whereas that of Sc^{+3} are colourless. Why? [1]
- b. Electoral stain ink typically contains a pigment for instant recognition, a silver nitrate. Why does silver nitrate produce black stain on skin? [1]

5. Zinc and Mercury are volatile metals which belong to group IIB in the periodic table.
- How base metals and oxide impurities are removed from impure mercury? [2]
 - How does white vitriol react with excess NaOH? [1]
 - Write the action of heat on white vitriol. [1]
 - Write the molecular formula of Nessler's reagent with its important use. [1]
6. Give reasons for
- $-\text{NO}_2$ group in nitrobenzene is meta director towards electrophilic substitution reaction. [2]
 - Ethoxyethane is stored in dark brown bottle containing iron wire. [2]
 - Phenol is more acidic than alcohol. [1]
7. a. Write down the suitable chemical reaction for the preparation of
i. 2-methoxypropane ii. methoxyethane, by using Williamson's synthesis. [2]
- b. Starting from CH_3ONa , how would you prepare methoxybenzene? What happens when methoxybenzene is treated with HI? [2+1]
8. a. Write reactions for
- Ethanol is heated with conc. H_2SO_4 at 140°C
 - Ethanol is refluxed with PCl_5 and the product is heated with alcoholic AgCN. [1+1.5]
- b. Give an example of each of the following reactions: Hydroboration oxidation [1]
- c. Explain, why are haloarenes less reactive towards nucleophilic substitution reactions? [1.5]

OR

Identify the major product, with name in the following

[5]



Group - C

Give long answer to the following questions.

[3 × 8 = 24]

9. a. Define: [2]
- i. Equivalent weight of acid
 - ii. Normality factor
- b. Write differences between equivalence point and end point. [2]

- c. 12 gram of commercial zinc is made to react with excess dil. H_2SO_4 . The total volume of H_2 gas was found to be 4.2 litres at 570 mmHg pressure and 279 K temperature.
- Calculate the mass of hydrogen gas produced at NTP. [2]
 - Determine the percentage purity of the zinc. [2]

OR

- a. Hydrogen and nitrogen oxide react according to the following equation:
 $2\text{H}_2(\text{g}) + 2\text{NO}(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g}) + \text{N}_2(\text{g})$

Experiments were performed at 800°C in order to determine the order of reaction and the following results were obtained.

Initial concentration of nitrogen oxide (mol litre^{-1})	Initial concentration of hydrogen (mole litre^{-1})	Initial rate of production of nitrogen ($\text{mol litre}^{-1} \text{sec}^{-1}$)
6×10^{-3}	1×10^{-3}	3×10^{-3}
6×10^{-3}	2×10^{-3}	6×10^{-3}
1×10^{-3}	6×10^{-3}	0.5×10^{-3}
2×10^{-3}	6×10^{-3}	2×10^{-3}

- What is the order of this reaction with respect to nitrogen oxide and hydrogen
 - Write the rate equation for the reaction of nitrogen oxide with hydrogen.
 - What is the rate of reaction if $[\text{H}_2] = 0.5\text{M}$ and $[\text{NO}] = 0.1\text{M}$ [2+1+1]
- b. Most common commercial batteries include Leclanche cell, lead storage accumulator and fuel cell like hydrogen-oxygen fuel cell.
- Why is MnO_2 and ZnCl_2 used in Leclanche cell? [1]
 - Name the anode and cathode used in the Lead storage cell. [1]
 - What is fuel cell? Write its net cell reaction. Mention any two advantages of the cell. [2]

10. Iron and aluminium are the most-used industrial metals in the world. Then comes copper.
- How is steel manufactured by Bessemer process? [3]
 - Mention any two methods of prevention of rusting. [1]
 - Write the reaction of copper with conc. HNO_3 . [1]
 - Write down the reactions involved in smelting process in the extraction of copper with well labeled diagram. [2]
 - Write the reaction involved in the preparation of Schweitzer's reagent starting from copper sulphate. [1]
11. Isopropyl alcohol is one of the ingredients of alcohol-based sanitizer and is a good disinfectant.
- Write the structure and IUPAC name of its functional isomer
 - How would you prepare it starting from 1- bromopropane?
 - How would you distinguish it from propan-1-ol?
 - What happens when isopropyl alcohol is oxidized?
 - What are
 - Rectified spirit?
 - Denaturated alcohol? [1+2+1.5+1.5+2]

OR

- A chloroalkane 'W' reacts with aq. KOH to produce 'X'. 'X' produces H_2 gas and compound 'Y' with sodium metal. 'Y' when reacts with iodoethane produces compound 'Z'. If compound 'X' produces trichloromethane when distilled with bleaching powder water paste, identify W, X, Y and Z. Also write reactions involved.
- Convert chlorobenzene into
 - DDT
 - Toluene
 - Biphenyl

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